Chapter 4 — Biggest obstacles and risk register

Scoring: **Likelihood (L)** and **Impact (I)** on a 1–5 scale, **Score = L × I**. Review monthly; anything at 15 or above goes on the management agenda.

4.1 The five obstacles that decide success

1. **Supply density, 24×7.** A customer stuck at 2 a.m. in Dwarka needs a truck *near Dwarka at 2 a.m.* Arrival time is the product. With too few partners online, the dispatcher sees `NO_DRIVER_FOUND`, and one bad night can lose a customer for life. *Answer:* launch in a limited set of zones, guarantee minimum earnings for night partners in the pilot, and expand only when median arrival stays under 35 minutes.
2. **Customer acquisition cost vs. low frequency.** People need a tow rarely, perhaps once every few years. Paid search for "towing near me Delhi" is expensive, and our contribution is about ₹210 per job (chapter 6). *Answer:* SEO (locality pages are already built), partnerships with dealers, insurers, fleets, parking operators and housing societies, and memberships that turn a one-off into a recurring customer.
3. **Leakage (disintermediation).** After one job, the customer has the driver's number and may call directly next time. *Answer:* masked calling, platform-only benefits (invoice, damage cover, tracking, membership), a partner agreement clause against solicitation (already in clause 3.4), and making the platform the partner's best source of jobs so leaving costs them more than staying.
4. **Trust and damage claims.** One scratched luxury car or one rude driver can go viral. *Answer:* KYC'd partners, trip PIN (built), photo condition report at pickup and drop (to build, AI-assisted in chapter 5), mandatory partner insurance, a fast claims SOP, and ratings with automatic suspension.
5. **Incumbent B2B lock-in.** Most new cars and insured cars already come with "free" RSA from Allianz, Europ, TVS Auto Assist and others. *Answer:* sell to these companies as their Delhi fulfilment partner instead of fighting them, and win uncovered customers: older vehicles, two-wheelers, commercial vehicles, and people whose RSA failed them.

4.2 Risk register

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R1	Not enough partners online in a zone or hour; long arrivals	Operations	4	5	**20**	Zone-limited launch; night incentives; fleet-company partners; demand heatmaps (AI-4)	Ops lead	Time-to-assign > 5 min; `NO_DRIVER_FOUND` > 5 %
R2	CAC higher than contribution per job	Commercial	4	4	**16**	SEO, B2B channels, memberships, referral credits; stop any paid channel with CAC > 1.5× contribution	Founder	Blended CAC > ₹300
R3	Customers or partners transact off-platform	Commercial	4	4	**16**	Masked calls, platform-only value, anti-solicitation clause, quality bonuses	Founder	Repeat rate falling while partner job count steady
R4	Partners don't remit commission and GST on cash jobs	Financial	3	4	12	Credit limit per partner; offset against online earnings; auto-pause above a balance; push online payment	Finance	"Providers owe platform" balance > 14 days
R5	Customer vehicle damaged; claim exceeds partner's means	Legal/Trust	3	5	**15**	Partner insurance mandatory; photo evidence; claims SOP; platform cover from year 2	Ops lead	Any claim > ₹25,000

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R6	Safety incident (assault, theft, accident during tow)	Safety	2	5	10	KYC and police verification of drivers, trip PIN, live tracking, SOS contact, suspension policy	Ops lead	Any incident
R7	Regulatory change: aggregator licensing, social-security contribution, DPDP, e-commerce amendments	Legal	3	3	9	Compliance calendar (chapter 3); budget 1 % of turnover for social security; counsel on retainer	Founder	Gazette notifications
R8	Wrong GST/TDS/TCS model causes back taxes and penalties	Tax	3	4	12	Decide principal vs. marketplace with CA before launch; monthly reconciliations	Finance + CA	CA review findings
R9	Personal-data breach	Security	2	5	10	Security baseline (`docs/developer/SECURITY.md`), CERT-In incident plan, least-privilege admin roles, log retention	Eng lead	Unusual admin logins; rate-limit spikes
R10	OSM public endpoints throttle us (geocoding/routing)	Technology	4	3	12	Self-host Nominatim/OSRM, or switch to Google/Mappls (provider interface already exists)	Eng lead	429 errors from geo API
R11	SMS OTP delivery failures (DLT, operator filtering)	Technology	3	4	12	DLT registration; Firebase phone auth with email fallback (built); WhatsApp OTP later	Eng lead	OTP verify rate < 85 %
R12	Polling load grows with live tracking at scale	Technology	2	3	6	SSE/WebSockets + Redis for locations (scaling path in ARCHITECTURE.md)	Eng lead	DB CPU > 60 % sustained
R13	Fraud: fake bookings, GPS spoofing, collusive fake jobs to earn incentives	Fraud	3	3	9	OTP-verified accounts; PIN-gated start; GPS sanity checks; anomaly detection (AI-7)	Eng + Ops	Completed jobs with zero movement
R14	Key-person dependency (solo founder, single developer)	Organisation	4	4	**16**	Documentation (this pack), second engineer, shared credentials vault, runbooks	Founder	—
R15	Funding gap before break-even	Financial	3	5	**15**	Lean pilot first; 18–24 months runway at seed; monthly burn review	Founder	Runway < 9 months
R16	Brand/trademark conflict on "RoadSaathi"	Legal	2	3	6	Trademark search before marketing spend; brand is a single setting to change	Founder	Search result
R17	Accounting or governance failure (GoMechanic case)	Governance	1	5	5	Ledger is source of truth; monthly close; external audit; no booking of unrealised revenue	Finance	Ledger vs. bank mismatch
R18	24×7 desk burnout and attrition	People	3	3	9	Shift rota, automation of routine steps, clear escalation, fair pay	Ops lead	Missed SLAs at night

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R19	Weather surges (monsoon waterlogging, fog season) overwhelm supply	Operations	4	3	12	Seasonal partner standby agreements; surge cap ≤ 1.5× with clear display; pre-positioning	Ops lead	IMD alerts
R20	A big B2B client takes most of the volume (concentration, as at Urgently)	Commercial	2	4	8	No single client > 30 % of revenue; keep the consumer channel	Founder	Concentration report

4.3 Top risks heat view

Impact ↓ / Likelihood →	1	2	3	4	5
5	R17	R6, R9	R5, R15	R1	
4		R20	R4, R8, R11	R2, R3, R14	
3		R12, R16	R7, R13, R18	R10, R19	